

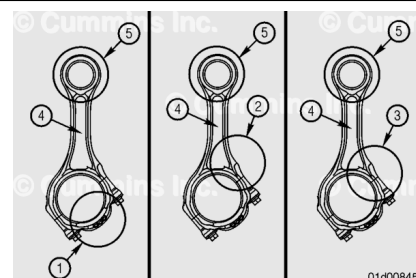
## Trainings ▾

### General Information

Different types of connecting rods are used in B series engines, depending on the producing plant and when the engine was built. **Not** all connecting rods are interchangeable. When replacing a connecting rod, use the following procedure for connecting rod identification and interchangeability. Refer to Procedure 001-999 in Section F.

(/qs3/pubsys2/xml/en/procedures/100/100-001-999.html)

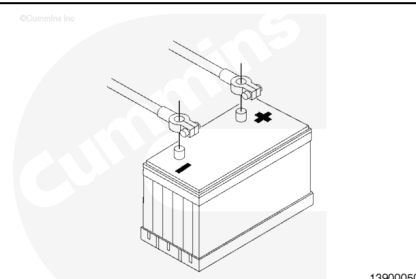
1. Balance weight on the connecting rod cap
2. Protrusion on the short side of the connecting rod
3. Smooth edge on the short side of the connecting rod
4. I-beam design
5. Oil grooves and thickness of connecting rod bushing.



### Preparatory Steps

#### **⚠ WARNING ⚠**

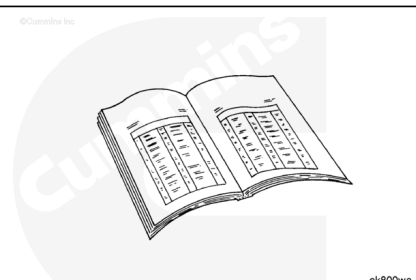
**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



Disconnect the batteries.

#### **⚠ WARNING ⚠**

**Coolant is toxic. Keep away from pets and children. If not reused, dispose of in accordance with local environmental regulations.**



#### **⚠ WARNING ⚠**

**Wait until the temperature is below 50°C [120°F] to reduce the possibility of personal injury from hot coolant.**

- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html>)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/40/40-002-004-tr.html>)

** WARNING **

**Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.**

** WARNING **

**To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.**

- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-037.html>)
- Remove the lubricating oil pan and gasket. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html>)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html>)
- If equipped, remove the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-089-tr.html>)
- Remove the J-jet piston cooling nozzles, if equipped, to prevent damage during connecting rod removal. Refer to Procedure 001-046 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-046-tr.html>)
- Remove the piston and connecting rod assemblies from the engine. Refer to Procedure 001-054 in Section 1.

(/qs3/pubsys2/xml/en/procedures/40/40-001-054.html)

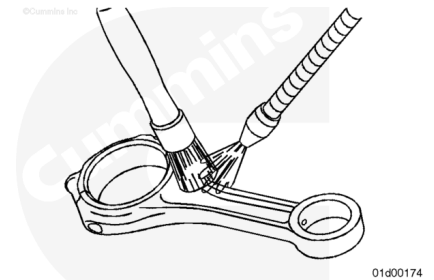
- Disassemble the pistons from the connecting rods. Refer to Procedure 001-054 in Section 1.

(/qs3/pubsys2/xml/en/procedures/40/40-001-054.html)

## Clean and Inspect for Reuse

### **⚠ WARNING ⚠**

**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**



### **⚠ WARNING ⚠**

**Compressed air used for cleaning should not exceed 207 kPa [30 psi]. Use only with protective clothing, goggles/shield, and gloves.**

### **⚠ CAUTION ⚠**

**The contact surface between the connecting rod and connecting rod cap is not a machined surface. Care should be exercised so these parts are not damaged during handling.**

Use solvent to clean the connecting rods.

**Note :** Unless cleaning the contact surface between the connecting rod and connecting rod cap, the rod and cap should be assembled together with the bolts tightened finger-tight to prevent damage to the mating surfaces.

Use solvent and a nylon bristle brush to clean the contact surface between the connecting rod and connecting rod cap.

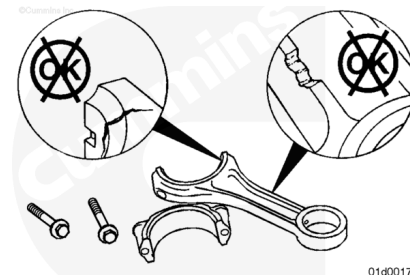
Dry the contact surface with compressed air.

Dry the connecting rods with compressed air.

### **⚠ CAUTION ⚠**

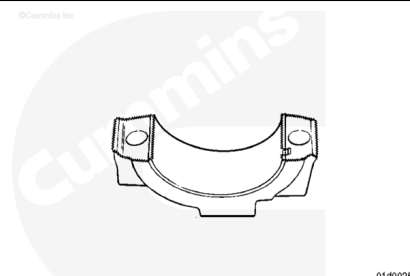
**The connecting rod and rod cap must be replaced as an assembly if any fretting damage is visible on either piece.**

Inspect the rod caps, connecting rod bearing saddles, and capscrews for nicks, cracks, burrs, scratches, and frets.



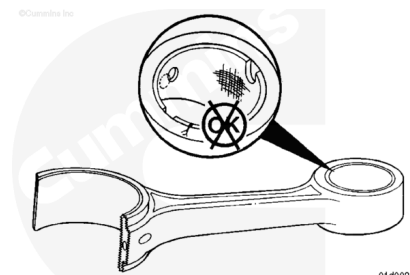
**Note :** On engines equipped with fracture-split connecting rods, the contact surface between the connecting rod and the connecting rod cap is **not** a machined surface. This surface should **not** be considered damaged unless there are deep nicks or cracks across the surface.

Inspect the contact surface between the connecting rod and the connecting rod cap.



Inspect the connecting rod piston pin bore for damage.

**Note :** Depending on when the engine was built, the connecting rod may or may not have an oil drilling in the top of the connecting rod small end. If replacing the bushing, the hole in the new bushing will **not** line up with the hole in the connecting rod. The hole in the bushing is for manufacturing purposes only and is no longer required to line up with the oil drilling in the connecting rod.



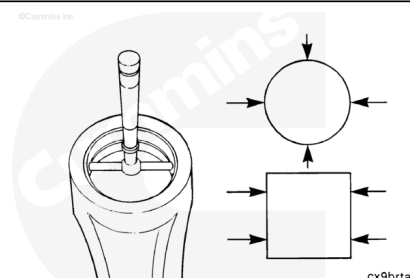
Measure the connecting rod piston pin bushing inside diameter.

B3.9, B4.5 RGT, and B5.9 Engines Connecting Rod  
Piston Pin Bushing Diameter

| mm     |     | in     |
|--------|-----|--------|
| 40.019 | MIN | 1.5756 |
| 40.042 | MAX | 1.5765 |

B4.5 Engines Connecting Rod Piston Pin Bushing  
Diameter

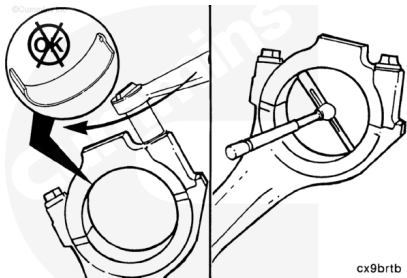
| mm    |     | in     |
|-------|-----|--------|
| 34.51 | MIN | 1.3586 |
| 34.54 | MAX | 1.3596 |



If the bushing is found to be damaged or out of specification, the bushing can be replaced. Use service tool, Part Number 3823902, to remove and install the bushing.

**⚠ CAUTION ⚠**

Use a vise with brass jaws to hold the connecting rod. Notches, scratches, or dents in the I-beam area can cause engine damage.



**⚠ CAUTION ⚠**

The number on the connecting rod must be the same as the number on the rod cap. Connecting rods and caps are manufactured as an assembly and cannot be interchanged. Engine damage can result.

Use clean 15W-40 engine oil to lubricate the connecting rod capscrews.

**Torque Value:**

B3.9, B4.5 RGT, and B5.9 Engines

- 1. 45 n•m [ 33 ft-lb ]
- 2. Rotate 90 degrees clockwise.

**Torque Value:**

B4.5 Engines

- 1. 25 n•m [ 18 ft-lb ]
- 2. Rotate 60 degrees clockwise.

Use a dial bore indicator to measure the inside diameter within a 20-degree arc from each side of the parting line.

Measure the inside diameter at 90 degrees from the parting line.

B3.9, B4.5 RGT, and B5.9 Connecting Rod Crankshaft Bore Diameter Bearings Removed

| mm    |     | in    |
|-------|-----|-------|
| 72.99 | MIN | 2.873 |
| 73.01 | MAX | 2.875 |

B4.5 Connecting Rod Crankshaft Bore Diameter Bearings Removed

| mm     |     | in     |
|--------|-----|--------|
| 69.675 | MIN | 2.7431 |

**B4.5 Connecting Rod Crankshaft Bore Diameter  
Bearings Removed**

| mm     |     | in     |
|--------|-----|--------|
| 69.694 | MAX | 2.7439 |

**Note : If the connecting rod crankshaft bore measurements are not within specifications, the connecting rod must be replaced.**

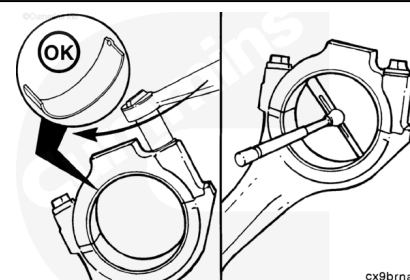
Repeat the above inspection with the connecting rod bearings installed.

**B3.9, B4.5 RGT, and B5.9 Standard Connecting Rod  
Crankshaft Bore Diameter with Bearings Installed**

|          | mm    |     | in    |
|----------|-------|-----|-------|
| Standard | 69.05 | MIN | 2.719 |
|          | 69.10 | MAX | 2.720 |

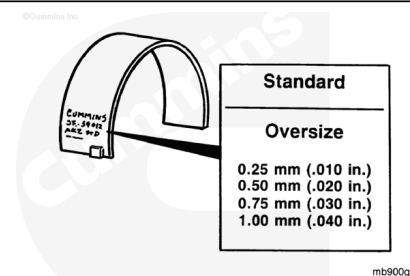
**B4.5 Standard Connecting Rod Crankshaft Bore  
Diameter with Bearings Installed**

|          | mm     |     | in     |
|----------|--------|-----|--------|
| Standard | 66.051 | MIN | 2.6004 |
|          | 66.103 | MAX | 2.6025 |



**B3.9, B4.5 RGT, and B5.9 Oversize Connecting Rod  
Crankshaft Bore Diameter with Bearings Installed**

|                       | mm    |     | in    |
|-----------------------|-------|-----|-------|
| 0.25 mm<br>[0.010 in] | 68.80 | MIN | 2.709 |
|                       | 68.85 | MAX | 2.711 |
| 0.50 mm<br>[0.020 in] | 68.55 | MIN | 2.699 |
|                       | 68.60 | MAX | 2.701 |
| 0.75 mm<br>[0.030 in] | 68.30 | MIN | 2.689 |
|                       | 68.35 | MAX | 2.691 |
| 1.00 mm<br>[0.040 in] | 68.05 | MIN | 2.680 |
|                       | 68.10 | MAX | 2.681 |



**Standard**

**Oversize**

0.25 mm (.010 in.)  
0.50 mm (.020 in.)  
0.75 mm (.030 in.)  
1.00 mm (.040 in.)

**B4.5 Oversize Connecting Rod Crankshaft Bore  
Diameter with Bearings Installed**

|                       | mm     |     | in     |
|-----------------------|--------|-----|--------|
| 0.25 mm<br>[0.010 in] | 66.801 | MIN | 2.6300 |
|                       | 66.853 | MAX | 2.6320 |
| 0.50 mm<br>[0.020 in] | 65.551 | MIN | 2.5809 |
|                       | 65.603 | MAX | 2.5828 |
| 0.75 mm<br>[0.030 in] | 65.301 | MIN | 2.5709 |
|                       | 65.353 | MAX | 2.5730 |
| 1.00 mm<br>[0.040 in] | 64.051 | MIN | 2.5587 |
|                       | 65.103 | MAX | 2.5608 |

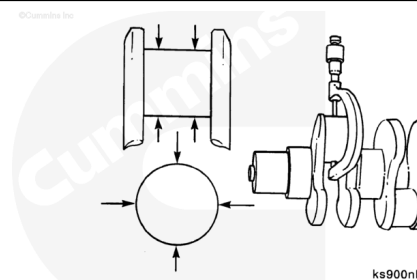
Measure the diameter of the connecting rod journal on the crankshaft.

**B3.9, B4.5 RGT, and B5.9 Standard Crankshaft  
Connecting Rod Journal Diameter**

|          | mm     |     | in     |
|----------|--------|-----|--------|
| Standard | 68.962 | MIN | 2.7150 |
|          | 69.013 | MAX | 2.7170 |

**B3.9, B4.5 RGT, and B5.9 Undersize Crankshaft  
Connecting Rod Journal Diameter**

|                       | mm     |     | in     |
|-----------------------|--------|-----|--------|
| 0.25 mm<br>[0.010 in] | 68.712 | MIN | 2.7052 |
|                       | 68.763 | MAX | 2.7072 |
| 0.50 mm<br>[0.020 in] | 68.462 | MIN | 2.6954 |
|                       | 68.513 | MAX | 2.6974 |
| 0.75 mm<br>[0.030 in] | 68.212 | MIN | 2.6855 |
|                       | 68.263 | MAX | 2.6875 |
| 1.00 mm<br>[0.040 in] | 67.962 | MIN | 2.6767 |
|                       | 68.013 | MAX | 2.6787 |



**B4.5 Standard Crankshaft Connecting Rod Journal  
Diameter**

|          | mm     |     | in     |
|----------|--------|-----|--------|
| Standard | 65.987 | MIN | 2.5979 |
|          | 66.103 | MAX | 2.5989 |

**B4.5 Undersize Crankshaft Connecting Rod Journal  
Diameter**

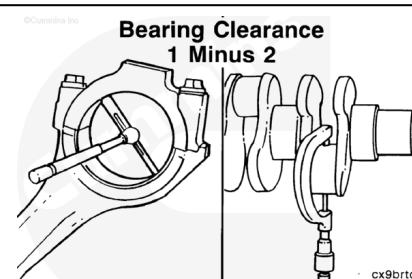
|                       | mm     |     | in     |
|-----------------------|--------|-----|--------|
| 0.25 mm<br>[0.010 in] | 65.737 | MIN | 2.5879 |
|                       | 65.763 | MAX | 2.5889 |
| 0.50 mm<br>[0.020 in] | 65.487 | MIN | 2.5779 |
|                       | 65.513 | MAX | 2.5789 |
| 0.75 mm<br>[0.030 in] | 65.237 | MIN | 2.5679 |
|                       | 65.263 | MAX | 2.5689 |
| 1.00 mm<br>[0.040 in] | 64.987 | MIN | 2.5579 |
|                       | 65.013 | MAX | 2.5589 |

**Note :** If crankshaft connecting rod journals are **not** within the given specifications, the crankshaft must be reground.

Bearing clearance equals the connecting rod crankshaft bore (with bearing) minus the crankshaft connecting rod journal diameter.

**Connecting Rod to Crankshaft Bearing Clearance**

| mm   |     | in    |
|------|-----|-------|
| 0.04 | MIN | 0.002 |
| 0.12 | MAX | 0.005 |

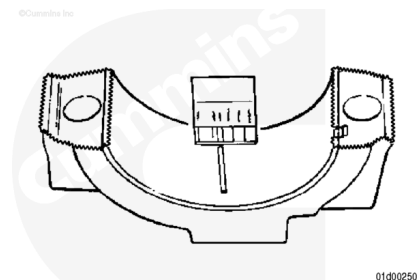


Bearing clearance can also be determined with a plastigauge during engine assembly.

**Note :** If the clearance is found to be out of specification, replace or try another set of connecting

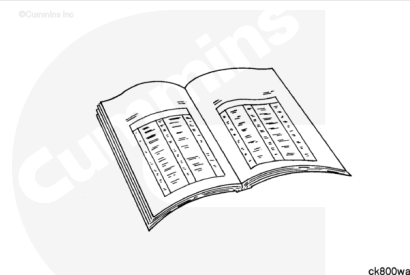


rod bearings.



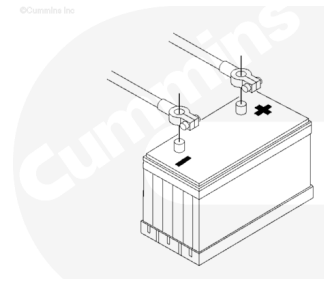
## Finishing Steps

- Install the pistons on the connecting rods. Refer to Procedure 001-054 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/40/40-001-054.html)
- Install the pistons and connecting rod assembly. Refer to Procedure 001-054 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/40/40-001-054.html)
- Install the J-jet piston cooling nozzles, if equipped. Refer to Procedure 001-046 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/40/40-001-046-tr.html)
- If equipped, install the block stiffener plate. Refer to Procedure 001-089 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/40/40-001-089-tr.html)
- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.  
(/qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html)
- Install the lubricating oil pan and gasket. Refer to Procedure 007-025 in Section 7.  
(/qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/40/40-002-004-tr.html)
- Fill the lubricating oil pan. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/40/40-007-037.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html).



**⚠ WARNING ⚠**

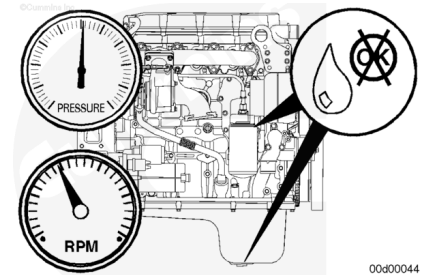
**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



13900050

Connect the batteries. See equipment manufacturer service information.

Operate the engine to normal operating temperature and check for leaks.



00300044